

Jaiveer Bassi

707-656-6227 | jaiveerbassi@yahoo.com | [linkedin.com/in/jaiveer-bassi](https://www.linkedin.com/in/jaiveer-bassi) | github.com/imjbassi | imjbassi.github.io

Experience

Physical Intelligence

May 2026 – Present

AI Robot Operator

San Francisco, CA

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through diverse manipulation, sorting, and assembly tasks with precision and consistency.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.

Handshake AI

Apr 2026 – Jun 2026

Contracted AI Response Evaluator

Remote

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

Kaiser Permanente Regional Laboratory

Oct 2023 – Jun 2024

Software Developer

Berkeley, CA

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

Projects

CAN Bus Fault-Injection & Diagnostics Rig | *Wiring, SocketCAN, Multimeter Diagnostics, MCP2515*

- Assembled a three-node CAN network by hand (Arduino/MCP2515 nodes, USB-CAN interface, custom wiring harness) and deliberately induced bus-off, arbitration, connector, wiring, and termination faults to study real failure signatures.
- Diagnosed each fault using multimeter continuity checks alongside SocketCAN error-state analysis, then published a reproducible fault catalog mapping physical failure modes to observable diagnostic evidence.

GELLO-Style Leader-Follower Teleoperation Rig | *Mechanical Assembly, Servo Control, Calibration*

- Built a four-DOF leader-follower mechanism by hand for approximately \$30, machining and assembling linkages, wiring potentiometer-based joint encoders, and calibrating control mapping without inverse kinematics.
- Documented hardware assembly, alignment, and calibration procedures, including force-feedback and gravity-compensation tradeoffs of the low-cost mechanical design.

UR5e Fault Injection & Diagnostics Harness | *Electrical/Controller Diagnostics, RTDE, URSim*

- Built a control and fault-characterization harness against a production robot arm's controller interface, driving motion and logging synchronized joint telemetry to catch failures invisible to standard exception handling.
- Injected connection and controller-level faults to distinguish recoverable errors from failures requiring external process supervision, directly informing repair versus restart decisions.

Castline Studio | *Resin 3D Printing, Print Calibration, Post-Processing*

- Operate and maintain resin 3D printing equipment for a production e-commerce business generating \$2K+/month, including print calibration, resin handling, curing, and post-processing to hold consistent part tolerances across 100+ product listings.
- Built the supporting order-management platform (Next.js/TypeScript) with a custom WebAssembly STL parser that analyzes uploaded geometry client-side to calculate volume, dimensions, and print cost before a job is queued.

Custom Mechanical Keyboards | *Soldering, PCB Assembly & Rework, Hand Tools*

- Hand-built and modified numerous mechanical keyboards from bare PCBs, soldering switches, diodes, and controllers, and reworking joints and traces to fix wiring and connectivity faults.
- Developed practical proficiency in through-hole and surface-level soldering, continuity testing, and PCB-level troubleshooting through repeated hands-on builds.

Research Publications

Not All Bad Demonstrations Are Equally Bad | Aug 2026

- Studied how flaws in robot demonstration recordings affect a robot's ability to complete tasks after training, running 78 controlled tests to isolate which types of bad recordings hurt performance most.

Smoothness Ranks Skill, Not Success | Aug 2026

- Tested a popular assumption in robotics, that smoother robot arm movement means better-quality training data, and found it often does not predict actual task success.

Efficient Packaging and Deployment of AI Models for Edge Inference | Sep 2025

- Achieved up to 4x model-size reduction and 50–70% faster inference using quantization, pruning, and lightweight containerization, making AI models fast enough to run directly on robots and other small devices.

Technical Skills

Hardware & Diagnostics: Electrical/mechanical troubleshooting, wiring and harness assembly, multimeter diagnostics, actuator and motor control, schematic interpretation, soldering & PCB assembly/rework, fault injection, root cause analysis, telemetry analysis
Embedded & Control Systems: RTDE, URSim, CAN/SocketCAN, teleoperation, real-time control
Languages: Python, C++, SQL, Bash
Systems: Linux, Git, Docker, Kubernetes, Redis Streams, PostgreSQL, RESTful APIs, AWS

Education

Grand Canyon University <i>M.S. in Computer Software Engineering</i> • Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages. GitHub	May 2024 – Oct 2025 Phoenix, AZ
University of Silicon Valley <i>B.S. in Computer Science</i> • Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine, optimizing recorded audio and keyword-triggered animations for real-time performance. GitHub	May 2021 – Aug 2023 San Jose, CA

Certifications

HackerRank Software Engineer Certificate (Mar 2026)	Cisco Network Support and Security (Sep 2025)
IBM Artificial Intelligence (Aug 2025)	Stanford AI and Machine Learning in Healthcare (Jun 2025)
Google Foundations of Project Management (Sep 2024)	Google IT Support Professional Certificate (May 2021)